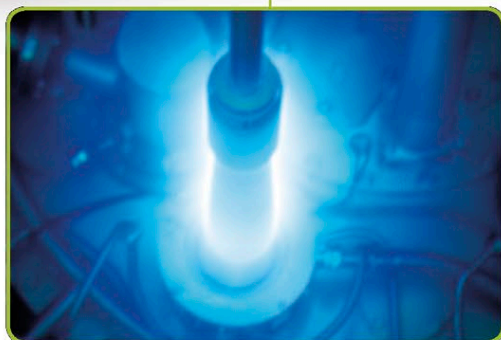
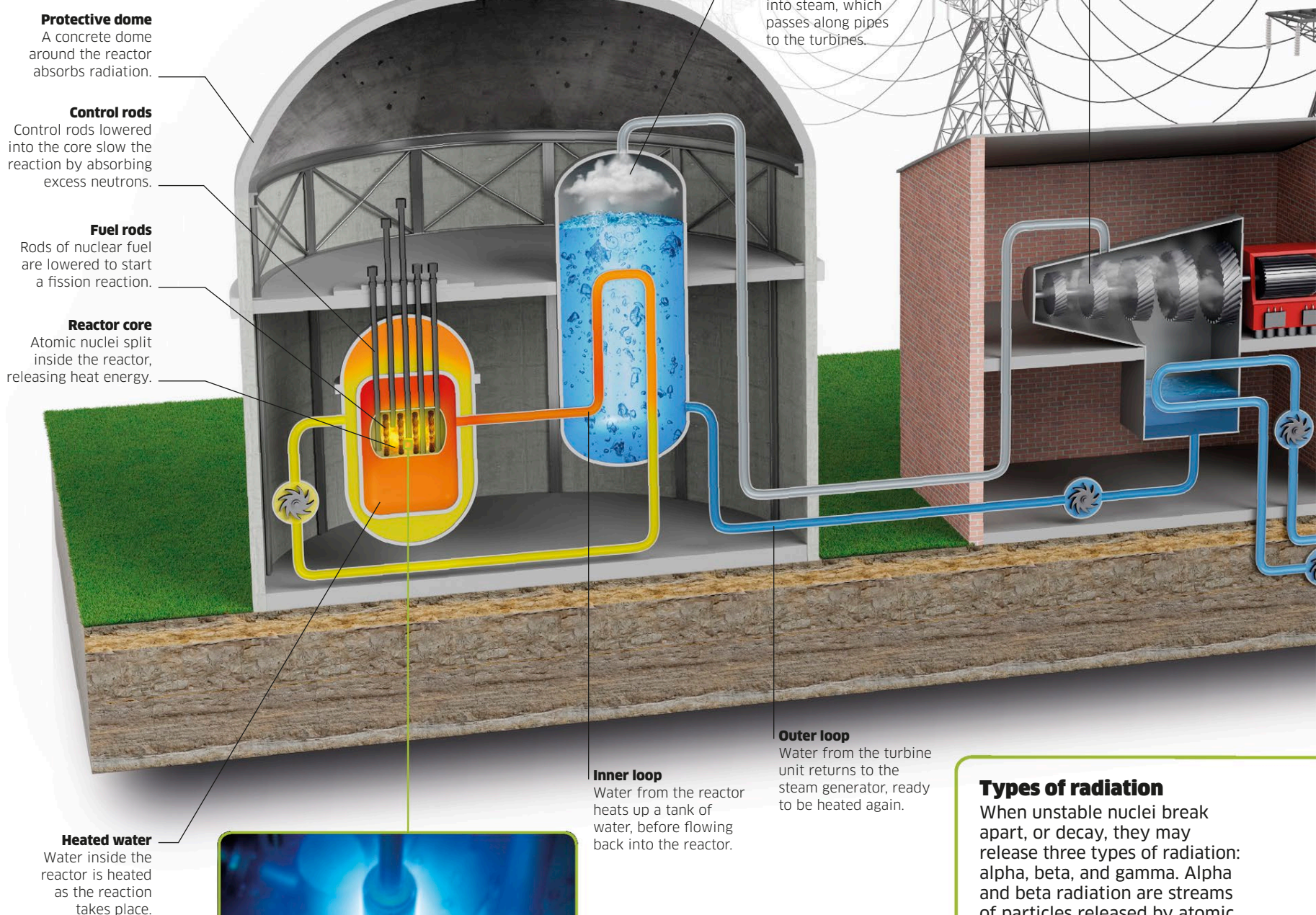


Nuclear energy

Nuclear reactions are a highly efficient way of releasing energy. Smashing atomic particles together sets off a chain reaction—producing enough heat to generate large amounts of electricity.

Most elements have several slightly different forms, called isotopes. Each isotope of an element has a different number of neutrons. Radioactive isotopes have too many or too few neutrons, making them unstable. Isotopes of heavy elements, such as uranium and plutonium, may break apart, or decay, producing radiation. Atomic nuclei can also be broken apart (fission) or joined together (fusion) artificially to release energy, which can be harnessed in nuclear power stations and weapons.



Cherenkov radiation

The atomic particles in the reactor travel incredibly fast. In doing so, they generate a type of radiation called Cherenkov radiation, which makes the water surrounding the reactor glow an eerie blue color.

Nuclear reactor

Nuclear fission power stations are found all over the world. They all use the same basic principles to generate electricity. Firstly, atoms are smashed apart in the reactor to release heat energy. This energy passes into a nearby chamber to heat up water and produce large quantities of steam. The steam powers spinning turbines attached to a generator, which converts this kinetic energy into the electricity that is pumped out to the world.

Types of radiation

When unstable nuclei break apart, or decay, they may release three types of radiation: alpha, beta, and gamma. Alpha and beta radiation are streams of particles released by atomic nuclei. Gamma rays, released during alpha and beta decay or even by lightning, are a form of electromagnetic radiation—similar to light, but more powerful and dangerous.